

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	02/07/2026
Team ID	
Project Name	Book Store
Maximum Marks	4 Marks

Technical Architecture Overview

The system architectural framework layout outlines the structural components, technical pipelines, interface paradigms, and database topologies supporting the ecosystem execution paths.

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Responsive digital marketplace storefront managing book search filters, inventory updates, and multi-step cart checkout flows.	React JS, Tailwind CSS, HTML5
2	Application Logic-1	Core service layer handling book processing workflows, catalog lookups, and transactional processing.	Python (FastAPI / Django)
3	Application Logic-2	Real-time book text scanner parsing printed book ISBN entries via optical camera ingestion.	Google Cloud Vision API
4	Application Logic-3	Automated book conversational assistant handling customer order status updates and tracking inquiries.	IBM Watson Assistant
5	Database	Relational transaction engine storing checkout logs, payment receipts, and tabular catalog edition records.	MySQL / PostgreSQL
6	Cloud Database	Distributed cache database service storing transient customer shopping session data.	IBM Cloudant / Redis
7	File Storage	Blob storage infrastructure hosting high-resolution bookstore catalog item cover graphics.	IBM Cloud Object Storage / AWS S3
8	External API-1	Retrieves precise real-time shipping tracking and delivery corridor validation checks.	FedEx / DHL Fulfillment API
9	External API-2	Verifies payment details and processes tokenized bank transactions securely.	Stripe / PayPal Gateway API
10	Machine Learning Model	Generates highly relevant multi-genre catalog cross-sell and up-sell recommendations based on reading historical logs.	Collaborative Filtering Model (Scikit-Learn)
11			

S.No	Component	Description	Technology
	Infrastructure (Server / Cloud)	Containerized production deployment framework ensuring platform stability and localized continuous delivery pipelines.	Kubernetes, Docker, Cloud Foundry

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Enables fast service routing, state persistence management, and responsive component assembly blocks.	FastAPI framework, React JS library
2	Security Implementations	Enforces data transit protection, application endpoint monitoring, and granular access token schemes.	SHA-256 Hashing, HTTPS TLS 1.3, Auth0 IAM
3	Scalable Architecture	Decoupled system architecture isolating search query indexing engines from heavy checkout processing servers.	Microservices Pattern, Docker Containers
4	Availability	Fault-tolerant execution patterns balancing incoming traffic loads across geo-distributed servers.	NGINX Load Balancers, Multi-Region Clusters
5	Performance	Optimizes webpage rendering speeds by serving catalog assets through edge networks and in-memory caches.	Redis Session Cache, Cloudflare CDN

References:

- <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>
- <https://www.ibm.com/cloud/architecture>